Arnav Pallem HW 3 Q4

4. $E[\hat{\beta}|X] = E\left(\frac{\sum_i Y_i U_i}{\sum_i U_i^2}\right) \approx \frac{E[\sum_i Y_i U_i]}{E[\sum_i U_i^2]} = \frac{\sum_i E[Y_i U_i]}{\sum_i E[U_i^2]}$

$E[Y_i U_i] = E[(\beta X_i + \epsilon_i)(X_i + \delta_i)] = E[\beta X_i^2 + \beta \delta_i X_i + \epsilon_i X_i + \epsilon_i \delta_i]$

$= \beta X_i^2 + 0$ (since all other random variables have 0 mean)

$E[U_i^2] = E[X_i^2 + 2X_i \delta_i + \delta_i^2] = X_i^2 + \tau^2$ $(V[\delta] = E[\delta^2] - E[\delta]^2)$

$\Rightarrow E[\hat{\beta}|X] = \frac{\beta \sum X_i^2}{\sum X_i^2 + \tau^2} = \boxed{\beta\left(\frac{\sum X_i^2}{\sum X_i^2 + n\tau^2}\right)}$

Since $\tau^2 > 0$, the measurement error will lead to a consistent underestimation of β. As n grows, the beta estimate goes to 0, the bias increases. As $\tau^2 \to \infty$, β will also go to 0, showing that the linear model will stop being a good model.